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Problem Solving and Design

TU914/1 Semester 2

Assignment 1 – Recycling Guide App

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Research

After doing some research on current world problems that we have now, I have noticed that pollution affects the world in many ways by bringing harmful materials into the environment (National Geographic, 2024). There are numerous causes of pollution and one of the major causes is incorrect waste disposal. Waste disposal is a key factor for many types of pollution such as water pollution, air pollution and soil pollution (United Nations Environment Programme, 2017).

When I did further research on this topic, I found out that water pollution was one of the most harmful forms of pollution to the environment and it can cause an algae bloom to occur in lakes or bodies of water which leads to eutrophication. Furthermore, harmful algae blooms can create neurotoxins that can harm the wildlife. Chemicals from incorrectly dumped waste by industrial and wastewater companies can have an enormous effect on the populace as the toxins can work their way up the aquatic food chain (Denchak, M., 2023). These effects of water pollution have a major impact on us because “According to the World Health Organization (WHO) survey, 80% of the world’s diseases and 50% of the world’s child deaths are related to poor drinking water quality” (Lin, L., Yang, H. and Xu, X., 2022).

Next, I found out more about air pollution and its effects caused by incorrect waste disposal. Some people may want to dispose their waste at home by using an incinerator however, this is illegal in many countries. It is a good regulation because burning waste produces a lot of toxic fumes and burning certain materials such as paper waste and menstrual products can release such toxins into the air as they may contain synthetic materials within them (Citizensinformation.ie, 2023). These harmful fumes can lead to poor air quality and thus can cause numerous adverse effects on the human body such as lung cancer and strokes (World Health Organisation, 2021).

Lastly, I did research on soil pollution and how incorrect waste disposal is a leading factor in the spreading. There are a number of soil pollutants that contribute to this problem such as heavy metals, plastic waste and toxic organic chemicals including pesticides. All three of these can come from incorrect disposal of waste (Münzel, T., Hahad, O., Daiber, A. and Landrigan, P., 2022). The soil will be heavily contaminated with heavy metals if wastewater is incorrectly managed and this can lead to an accumulation of these elements in the vegetables that may be harvested from that particular land. Likewise, incorrect use and

disposal of pesticides can lead to a very similar consequence as the toxic chemicals can remain in the soil and find their way into the vegetation that may be farmed there (Zwolak, A., Sarzyńska, M., Szpyrka, E. and Stawarczyk, K., 2019). Next, plastic waste being incorrectly dumped in soil can lead to releasing harmful chemicals into the surrounding soil and could have micro plastic seep into the soil thus, leading to micro plastic appearing within our food and water (UN Environment Programme, 2021).

In conclusion, I found out that improper waste disposal creates numerous issues for the environment, the plants and the animals. I decided that I should try solve a part of this problem by focusing my attention on how ordinary consumer like myself can dispose of their waste correctly. I went on to research the different types of bins and the materials that are associated with those bins. I want to create an app that can help the average consumer recognise what each product they buy should go into which bin. I understand that the biggest factor in pollution comes from incorrect waste disposal from large companies however I hope that starting with the average populace can lead to a meaningful change.

Business Case

App Idea

A mobile app idea that I would like to present is called the Recycling Guide App. It allows users to input the item they want to recycle, and they will be guided through the recycling process, along with explanations of why it should be recycled in that manner.

This app will assist and encourage people to recycle their waste, and to spread awareness regarding sustainability.

Target Audience

I would target this app towards children and parents more because I believe children should start recycling wastes starting from a young age. Thus, it is also crucial for those around them to act as a role model by setting good examples - the parents in this case. However, people from different age groups and occupations are also welcome to use this app.

Market Research

After doing some market research, I found a similar app called EcoScan (EcoScan app, 2019). Their app is provided on websites and mobile phones. They generate their revenue through a shop on the app that sells bins and related items. However, the explanations they provide are limited, unclear, and it was only shown after the user inputs the item they want to recycle.

Meanwhile, my app will have plenty of information and the explanations will be clear too, it is also provided in the home screen, in case the user wants to know more. It also offers features that attracts and encourage children to recycle more.

I think users would choose my app over the others because mine not only gives clear information and explanations, but it also looks attractive and fun to the children.

Monetization Strategy

This app will generate revenue through ads. The information this app provides will only be explained briefly, thus if the user wishes to learn more, the user must watch an ad to access the rest of the information.

App Objectives

The main goal of this app is to assist people with managing their waste, to spread awareness of sustainability, and to keep the environment healthy for future generations.

App Functionality

Features List

- Guide users on how they should recycle their waste.
- Provide explanations on why the user should recycle in that manner.
- Provide information regarding sustainability and waste management.
- Allow users to choose an animal character as a companion to accompany them throughout their recycling journey
- Allow users to choose the language they would prefer on the app.

User Interaction Flow

1. The first screen will ask the user if they would like to make an account or continue as a guest and to input the language they would like to be in the app. The next screen will ask the user if they would like to choose an animal character to be their companion in the app.
2. When the user finishes setting up, they will then be shown the home screen of the app where there will be 3 options for the user to choose, and that is if they would like to input the item that they want to recycle through description, barcode or photo.
3. After the user inputs an item they would like to recycle, a new screen will appear displaying instructions, explanations and information. However, the user can also choose to go back to the home page immediately without reading the rest of the text, and they can do this by pressing the icon at the top right corner of the screen.
4. Within the home screen, there is also an icon at the bottom of the screen that represents the waste management screen. This screen will offer the user 3 categories to choose from based on the types of bins. Then within these categories there will be information and explanations on what type of waste should go into each bin. However, if the user wishes to learn even more, they are required to watch an ad in order to get access to the rest of the information.
5. There is also an icon in the home screen that represents the settings screen, which allows the user to view or change the app settings and account. Through their account, they access the history of what they have recycled so far.

User Interface Design

Screen 1:

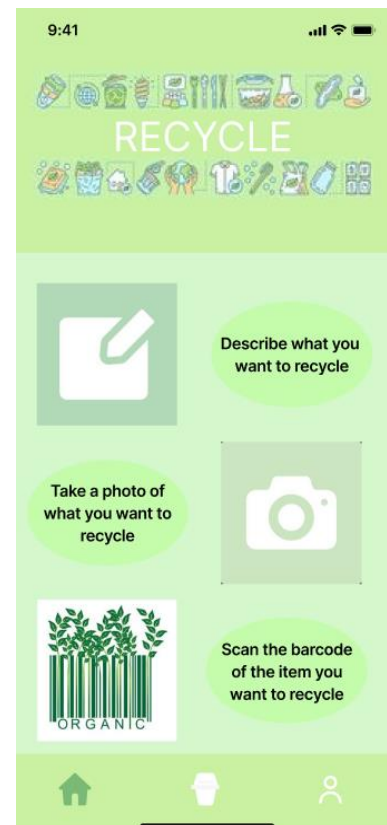
This screen will be the home page of the app

In the middle of the screen, there are words to indicate what each icon will allow the user to do.

- The first one allows the user to describe the item they want to recycle, thus why the icon is pen and paper.
- The second one allows the user to take a photo of the item they want to recycle, thus why the icon is a camera.
- The last one allows the user to scan the barcode of the item they want to recycle, thus why the icon is a barcode.

At the bottom of the screen, there will be 3 icons to indicate the main screens of the app and whichever screen the user is on, that icon of that screen would be of a darker colour.

- The left one indicates the home page.
- The middle one indicates the waste management screen.
- The right one indicates the settings screen.



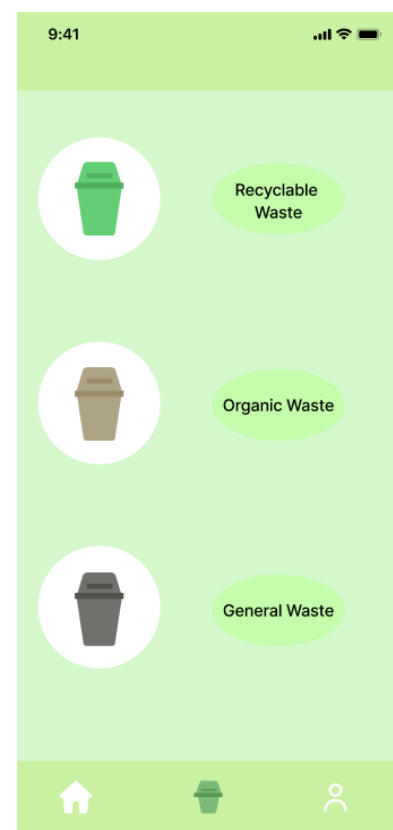
Screen 2:

This is the waste management screen, and it will be the first thing the user sees after pressing the waste management icon from the homepage screen.

This screen has 3 categories for the user to choose from.

- The first category has a green bin icon and words to indicate that it is for recyclable waste.
- The second category has a brown bin icon and words to indicate that it is for organic waste.
- The last category has a black bin icon and words to indicate that it is for general waste.

The reason I used those colours for each bin is because they indicate what waste should go in those bins.



User – Defined Classes

After some consideration, I think the key user – defined classes that the app would need to use are:

- Description
 - To store and record the description of the item the user has input.
 - The role of this class in this app would be taking in the user’s description input of the item they want to recycle.
 - Instance Variables
 - Sentence
 - Methods Headings
 - getSentence
 - setSentence
 - toString
- Photo
 - To store and record the photo of the item the user has taken.
 - The role of this class in this app would be taking in the user’s photo input of the item they want to recycle.
 - Instance Variables
 - Picture
 - Methods Headings
 - getPicture
 - setPicture
 - toString
- Code
 - To store and record the barcode of the item the user has scanned and input.
 - The role of this class in this app would be to take in the user’s barcode input of the item they want to recycle.
 - Instance Variables
 - Barcodes
 - Methods Headings
 - getBarcodes
 - setBarcodes
 - toString

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